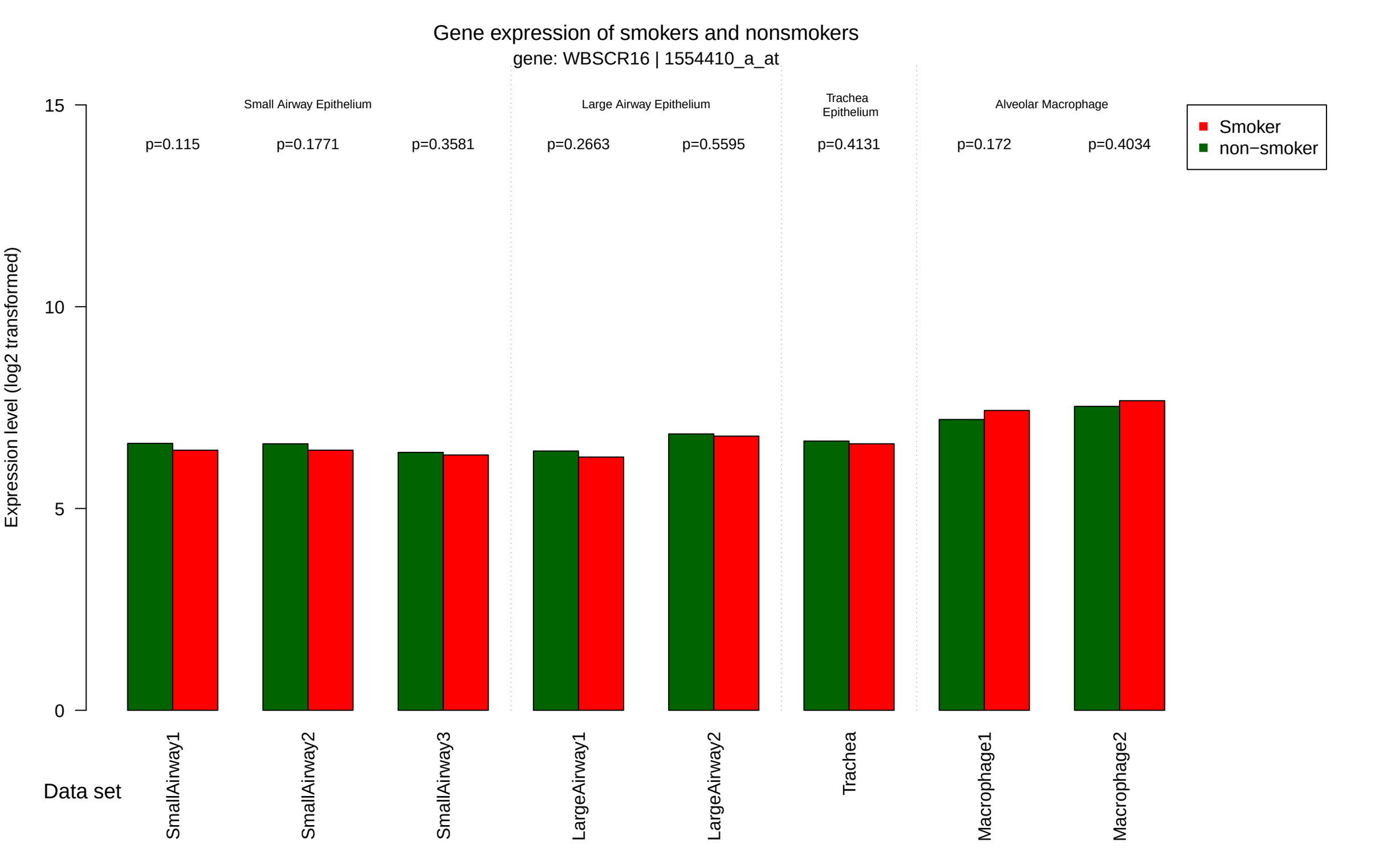
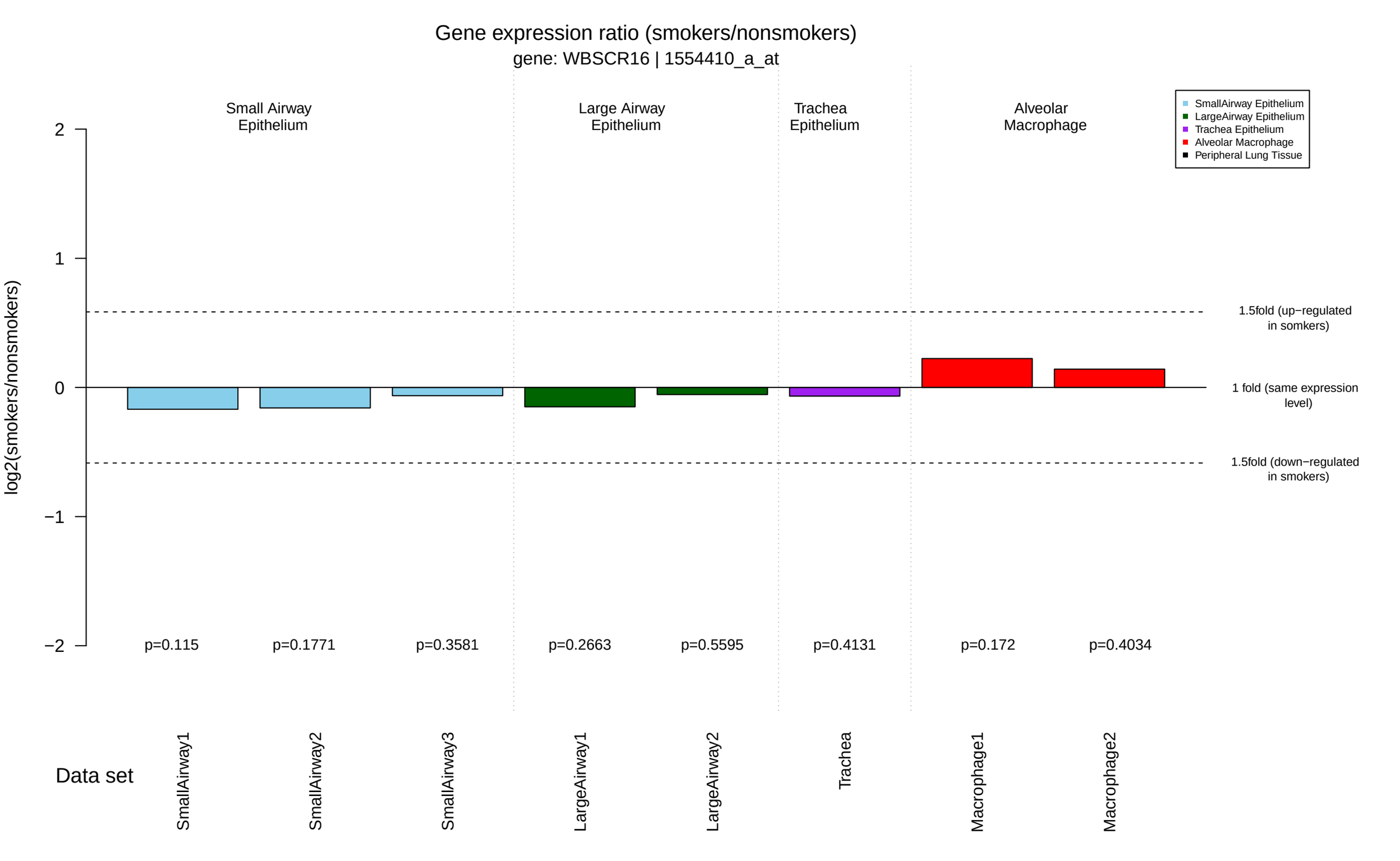
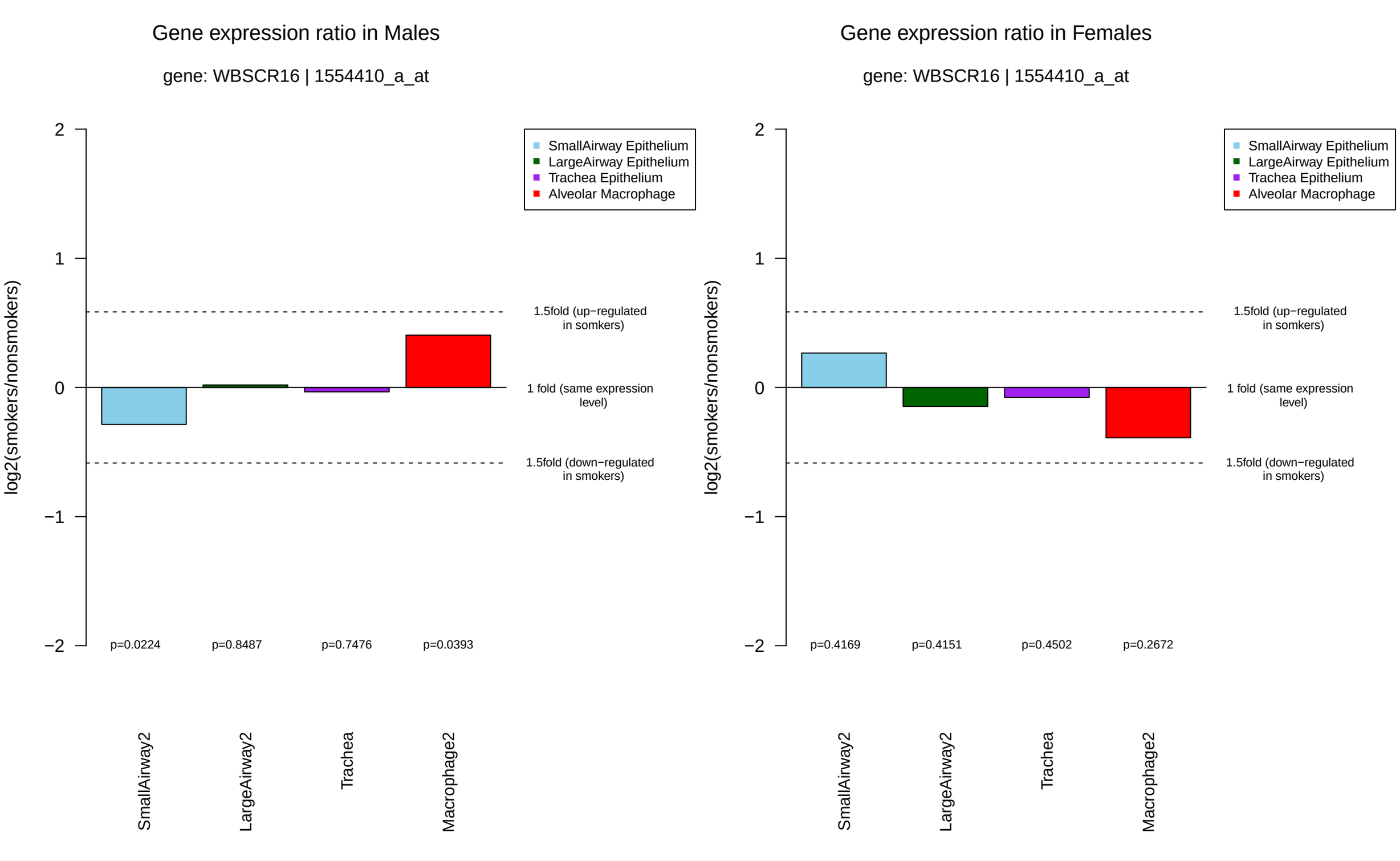
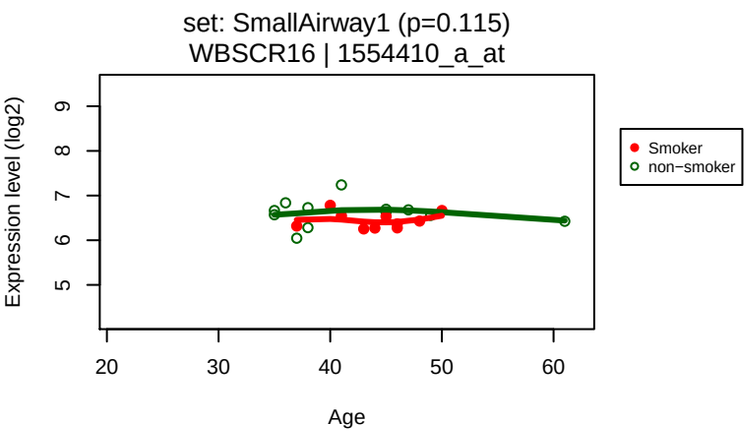
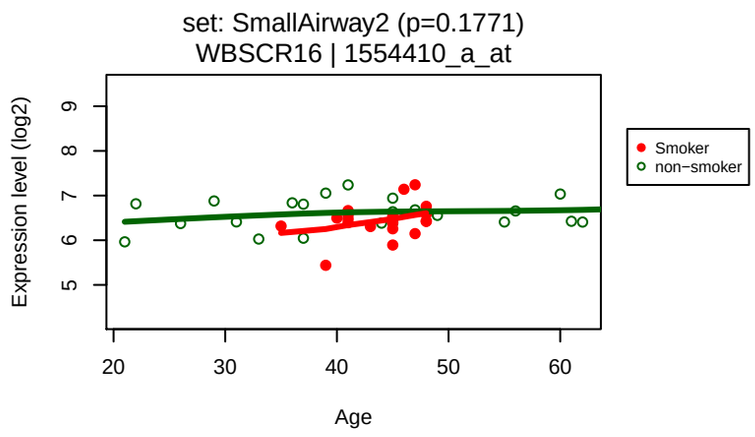
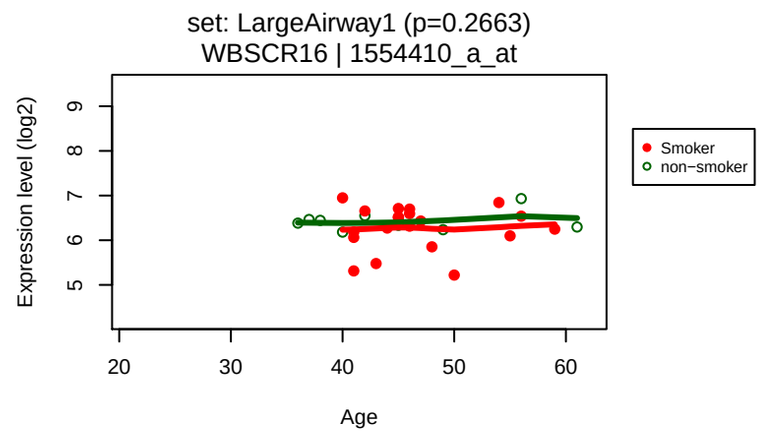

Gene expression of smokers and nonsmokers
gene: WBSCR16 | 1554410_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.115
p=0.1771
p=0.3581
p=0.2663
p=0.5595
p=0.4131
p=0.172
p=0.4034
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

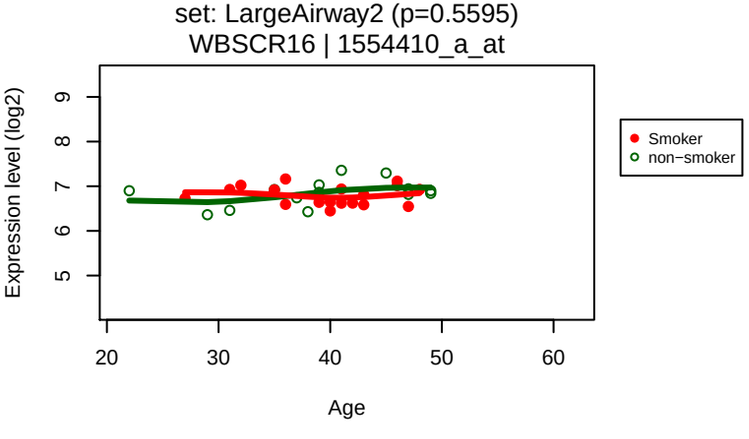

Gene expression ratio (smokers/nonsmokers)
gene: WBSCR16 | 1554410_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.115
p=0.1771
p=0.3581
p=0.2663
p=0.5595
p=0.4131
p=0.172
p=0.4034
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

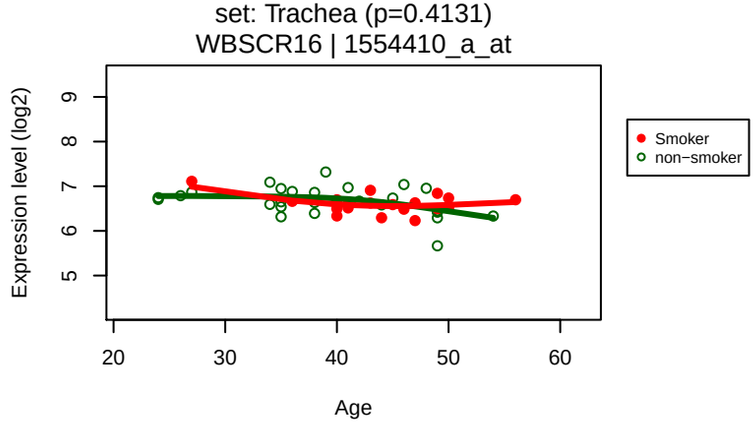

Gene expression ratio in Males
gene: WBSCR16 | 1554410_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0224
p=0.8487
p=0.7476
p=0.0393
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WBSCR16 | 1554410_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4169
p=0.4151
p=0.4502
p=0.2672
SmallAirway2
LargeAirway2
Trachea
Macrophage2

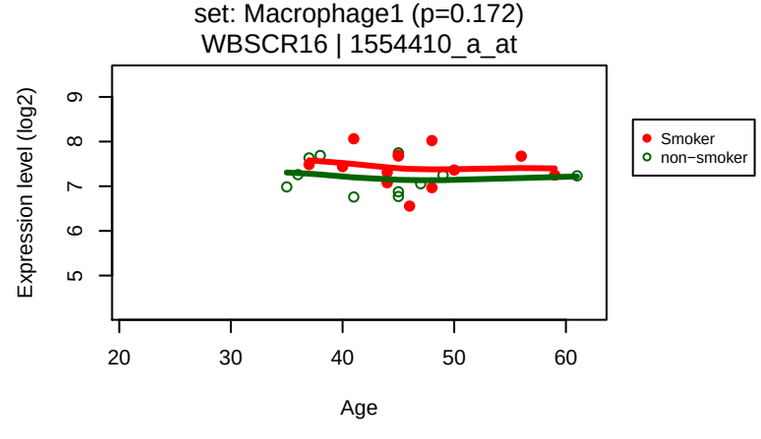

set: SmallAirway1 (p=0.115)
WBSCR16 | 1554410_a_at
Expression level (log2)
Age
Smoker
non−smoker

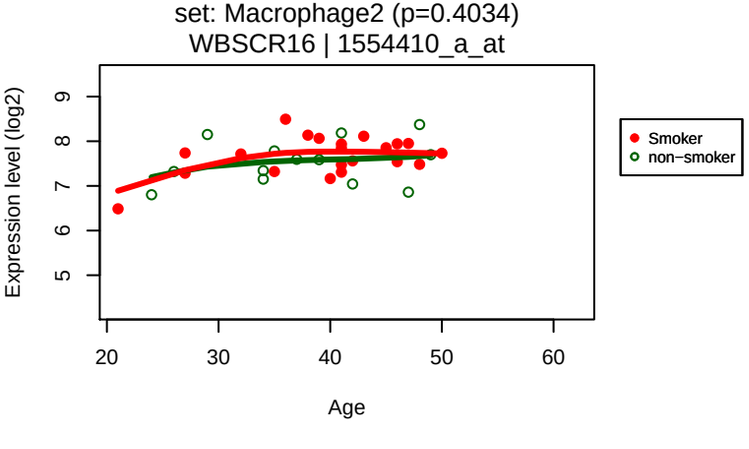

set: SmallAirway2 (p=0.1771)
WBSCR16 | 1554410_a_at
Expression level (log2)
Age
Smoker
non−smoker

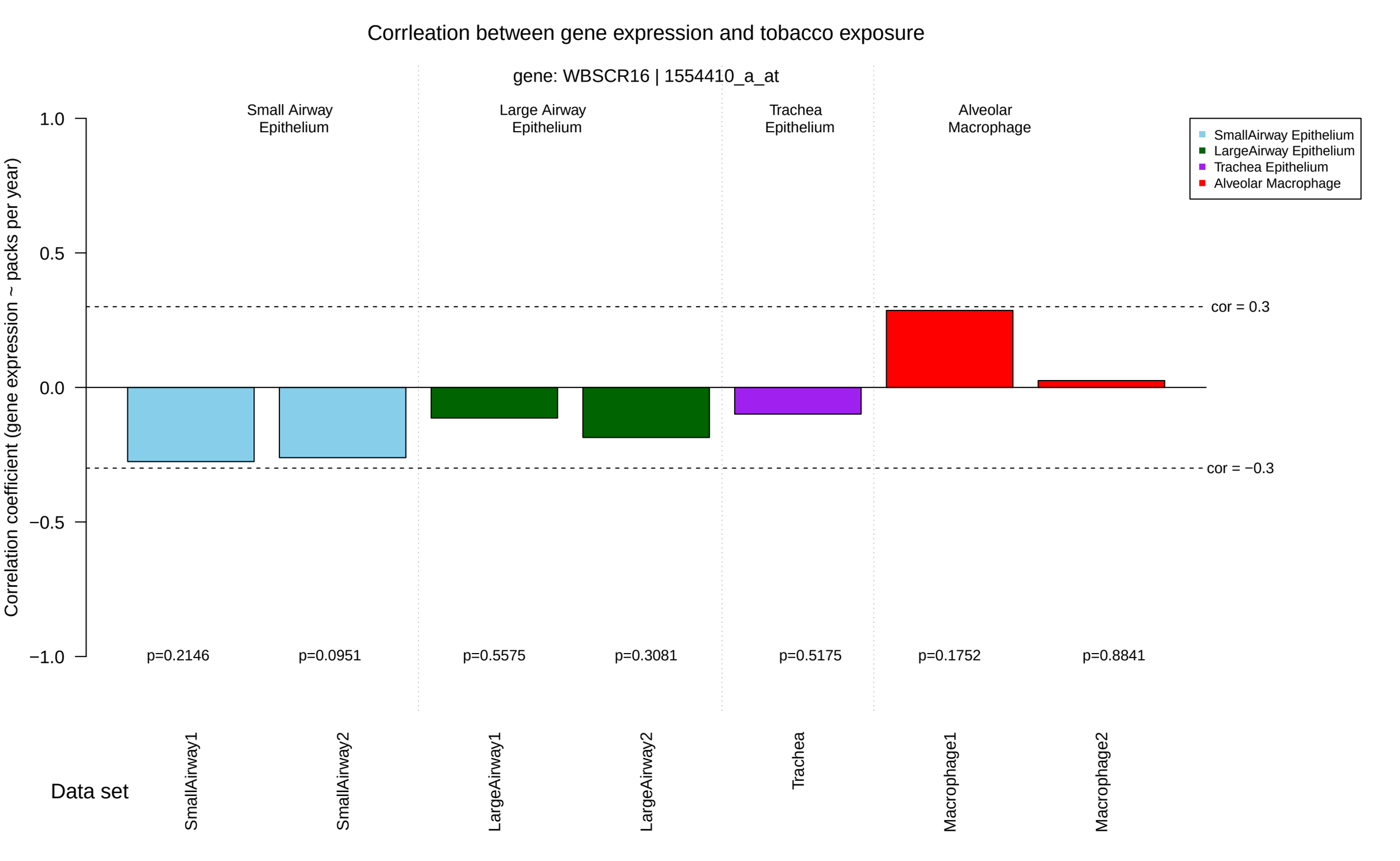

set: LargeAirway1 (p=0.2663)
WBSCR16 | 1554410_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5595)
WBSCR16 | 1554410_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4131)
WBSCR16 | 1554410_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.172)
WBSCR16 | 1554410_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4034)
WBSCR16 | 1554410_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WBSCR16 | 1554410_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2146
p=0.0951
p=0.5575
p=0.3081
p=0.5175
p=0.1752
p=0.8841
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2